

Fake News Detection

Detecting Truth in the Digital World



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Presentation Outline

Topic and Motivation

Dataset and Preprocessing

Exploratory Data Analysis

Machine Learning Experiments

Interactive Application

Conclusion

Topic and Motivation

Introduction

📑 Project Topic

Fake News Detection using Machine Learning.

Main Problems

- Fake news spreads rapidly online
- Influences public opinion
- Manual verification is difficult
- Automated systems are required

🎯 Goal

Develop a system capable of classifying news as Fake or Real.



Why Fake News Detection Matters

-  Elections and politics
-  Public health
-  Financial markets
-  Social stability
-  Public trust

Fake news often imitates professional journalism and becomes difficult to distinguish from real information.



Dataset and Preprocessing

Dataset Sources

Dataset	Purpose	Size
ISOT Fake News	Articles	44,898
LIAR Dataset	Statements	12,791
WELFake Dataset	Classification	72,134

129,823

Total Raw Samples

Data Cleaning Pipeline

Raw datasets contained noisy, duplicated, and inconsistent information. Preprocessing was required to improve dataset quality before training machine learning models.

1. Missing value removal
2. Duplicate removal
3. URL cleaning
4. HTML cleaning
5. Lowercase conversion
6. Stopword removal
7. Lemmatization

First 5 Rows:

	text	label	caps_ratio	exclamation_count	text_len	question_count	clean_text
0	WASHINGTON (Reuters) - The head of a conservat...	1.0	0.016000	0	4659	0	head conservative republican faction congress ...
1	WASHINGTON (Reuters) - Transgender people will...	1.0	0.011200	0	4077	0	transgender people allowed first time enlist m...
2	WASHINGTON (Reuters) - The special counsel inv...	1.0	0.015284	0	2789	0	special counsel investigation link russia pres...
3	WASHINGTON (Reuters) - Trump campaign adviser ...	1.0	0.010610	0	2461	0	trump campaign adviser george papadopoulos tol...
4	SEATTLE/WASHINGTON (Reuters) - President Donal...	1.0	0.017585	1	5204	1	president donald trump called postal service f...

TF-IDF Vectorization

TF-IDF (Term Frequency – Inverse Document Frequency) converts textual data into numerical vectors that machine learning models can process.

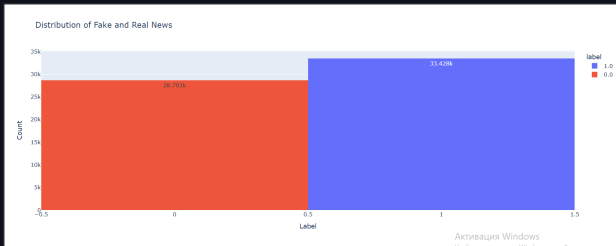
```
) from sklearn.feature_extraction.text import TfidfVectorizer  
  
tfidf = TfidfVectorizer(max_features=5000, stop_words='english')  
  
X_train_tfidf = tfidf.fit_transform(X_train)  
X_test_tfidf = tfidf.transform(X_test)
```

Explanation:

- TF measures how frequently words appear in articles
- IDF reduces the importance of overly common words
- Text data is transformed into sparse numerical vectors
- TF-IDF helps models identify meaningful vocabulary patterns

Exploratory Data Analysis

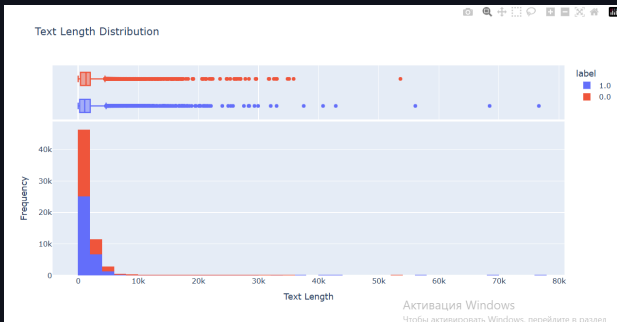
Class Distribution



Key Insights

- Balanced dataset
- Reduced risk of model bias
- Fair class representation
- More reliable model evaluation

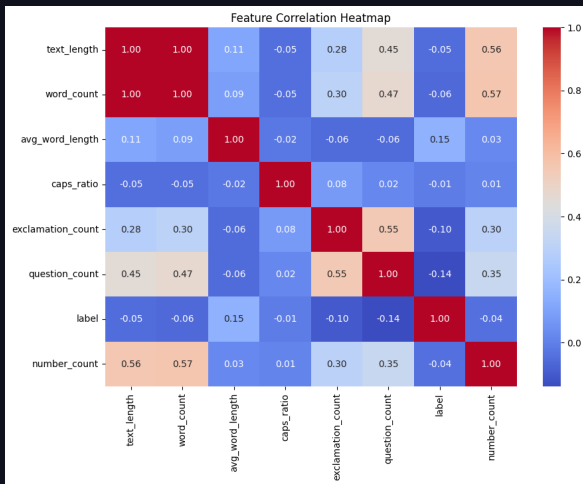
Text Length Distribution



Observations

- Fake articles vary more
- Real news is structured
- Text length is predictive
- Useful stylometric feature

Feature Correlation Heatmap

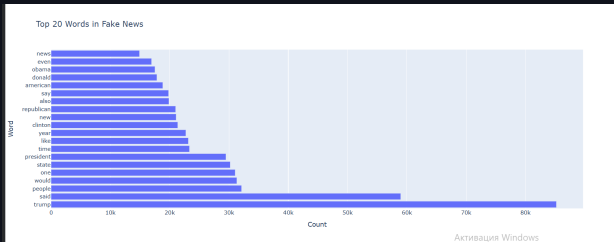


Interpretation

The heatmap shows relationships between engineered textual features.

- Strong correlation between text length and word count, other features show moderate correlations.
- No single feature strongly correlates with the label, indicating that fake news detection depends on a combination of multiple stylometric features.

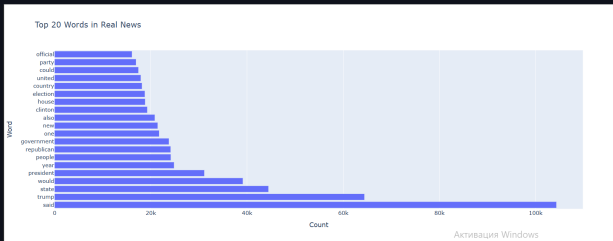
Top Words in Fake News



Observations

- Emotional vocabulary appears frequently
- Political and sensational terms dominate
- Repeated patterns help classification
- Fake news uses attention-grabbing language

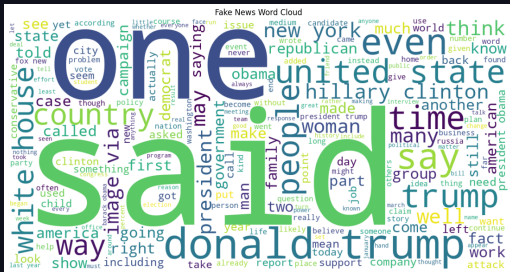
Top Words in Real News



Observations

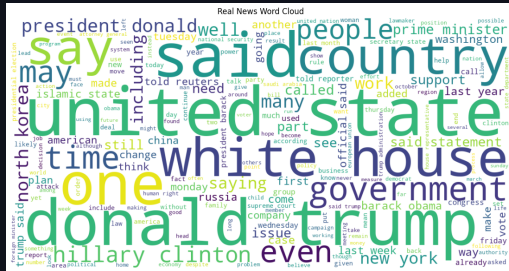
- Real news uses more formal vocabulary
- Informational language dominates
- Journalistic writing style is visible
- Less emotional wording appears

Word Clouds



Fake News

Fake news articles contain more emotional, sensational, and politically charged vocabulary. Repeated terms often aim to attract attention and provoke reactions.

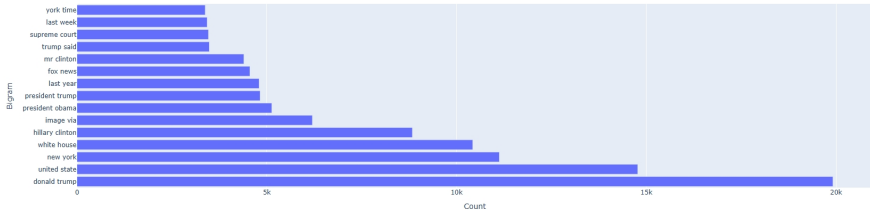


Real News

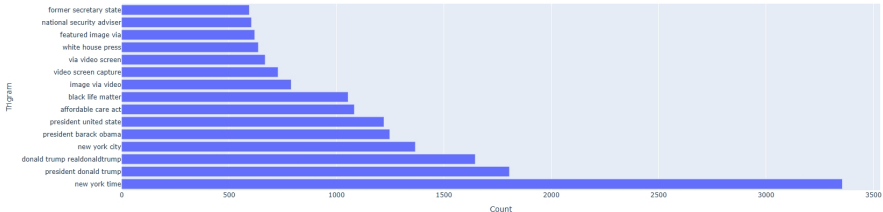
Real news articles use more formal, factual, and informational vocabulary. The language is generally more neutral and journalistic.

Top Fake News Bigram and Trigram Analysis

Top Fake News Bigrams



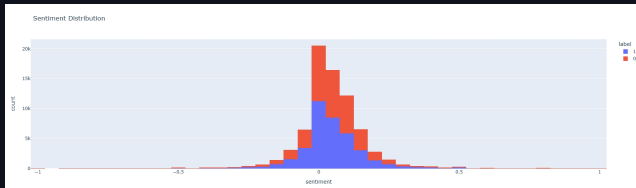
Top Fake News Trigrams



Bigram and Trigram Analysis

- Bigrams reveal recurring fake news topics related to U.S. politics and public figures such as Trump, Hillary Clinton, White House, and Obama.
- Political entities, locations, and repetitive phrase structures appear frequently in fake news content.
- Trigrams capture repetitive publishing patterns such as “image via video” and “video screen capture”, while real news articles less frequently use such repetitive template-like phrases.
- Political phrases such as “black lives matter” and “affordable care act” also appear frequently in fake news content.

Sentiment Distribution



Interpretation

- Large peak around 0 indicates most articles have neutral sentiment scores
- Fake and real news show overlapping distributions
- Slight shift toward positive sentiment values (moderately positive language appears more frequently than strongly negative language)

Machine Learning Experiments

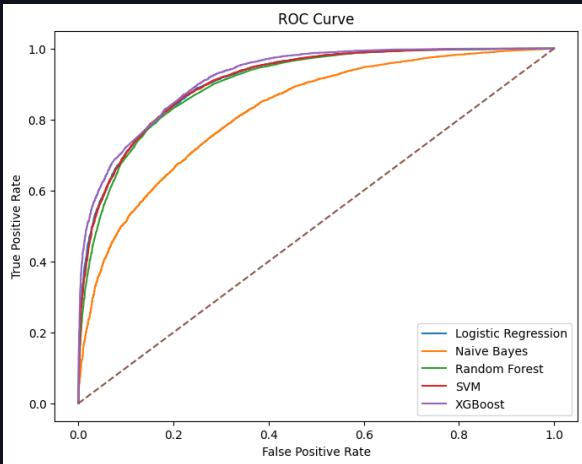
Models Used

- Logistic Regression
- Naive Bayes
- Random Forest
- SVM
- XGBoost

	Model	Accuracy	Precision	Recall	F1 Score
0	Logistic Regression	0.819356	0.812747	0.844564	0.828350
1	Naive Bayes	0.734036	0.756591	0.714542	0.734966
2	Random Forest	0.816309	0.820189	0.824926	0.822551
3	SVM	0.821675	0.816511	0.844179	0.830115
4	XGBoost	0.825782	0.787457	0.907329	0.843154

Multiple models were compared to identify the best classifier.

ROC Curve



Interpretation

- Curves close to the top-left corner indicate strong classification performance
- XGBoost and SVM achieved the best ROC performance
- Naive Bayes showed weaker class separation capability

Why Ensemble Learning Worked Best

- Combines strengths of multiple models
- Reduces weaknesses of individual algorithms
- Improves prediction stability
- Produces better generalization

ENSEMBLE

SVM + XGBoost

Final Model Results

Model	Acc	Prec	Recall	F1
Tuned XGBoost	0.834	0.807	0.909	0.855
Tuned SVM	0.860	0.863	0.879	0.871
Ensemble	0.863	0.857	0.894	0.8755

0.8755

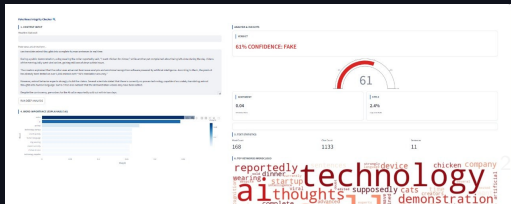
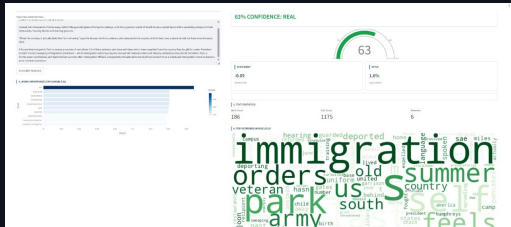
Best F1-score

Interactive Application

Interactive Prediction System

To demonstrate practical usability, an interactive prediction system was created. The application allows users to input a news article, receive prediction results, and classify the article as either Fake or Real.

1. User inputs article text
2. Text preprocessing
3. TF-IDF transformation
4. Ensemble prediction
5. Fake or Real result



Conclusion

Key Findings

- Fake news has measurable linguistic patterns
- TF-IDF is highly effective
- Stylometric features improve performance
- SVM and XGBoost produced strong results
- Ensemble achieved highest F1-score

Fake News Can Be Detected

Thank You!

Questions?

